

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks: 25

Annexure A

Comparative Analysis

Code of Conduct:

I Harish Kumar / 192521340 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Comparative Analysis

1) Star Topology vs. Bus Topology — Cost, Reliability, and Performance

Cost

Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub, plus a dedicated cable run from that device to every individual node. Bus topology, by contrast, uses a single shared backbone cable, so it needs far less cabling and no central device at all — making it the lower-cost, easier-to-install option for very small, budget-constrained networks.

Reliability

In terms of reliability, star topology is clearly superior: because each node has its own independent link back to the central switch, the failure of one node's cable affects only that node, leaving every other device fully connected. Bus topology has no such isolation — a break anywhere along the single shared backbone cable can bring down communication for every device on that segment, since they all depend on the same physical medium.

Performance

Performance also favours star topology: with a central switch managing traffic, each device effectively gets a dedicated collision-free path, so performance stays consistent even as more devices are added. Bus topology, however, degrades as more devices are attached, since all of them contend for the same shared medium — more devices directly means more contention, more collisions, and falling effective throughput.

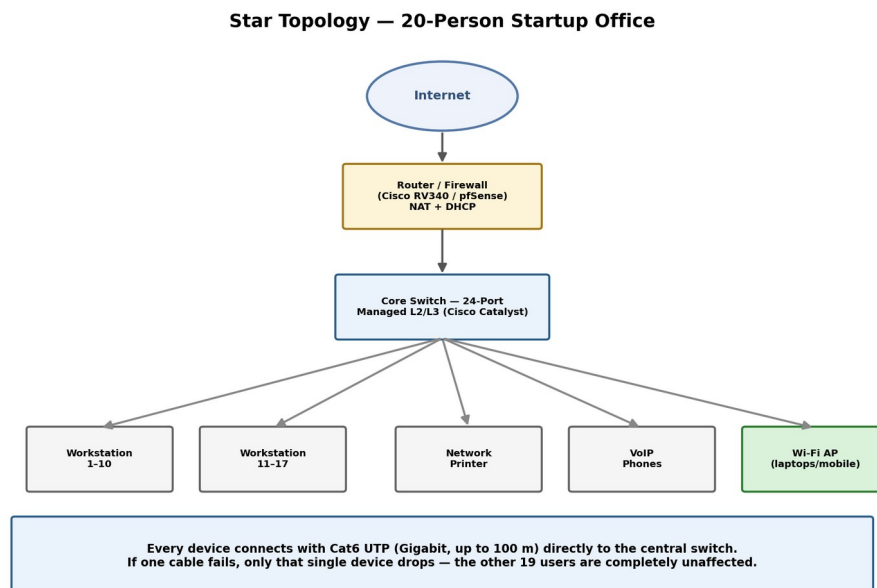


Figure 1.1 – Star topology: every device has its own dedicated link to a central switch.

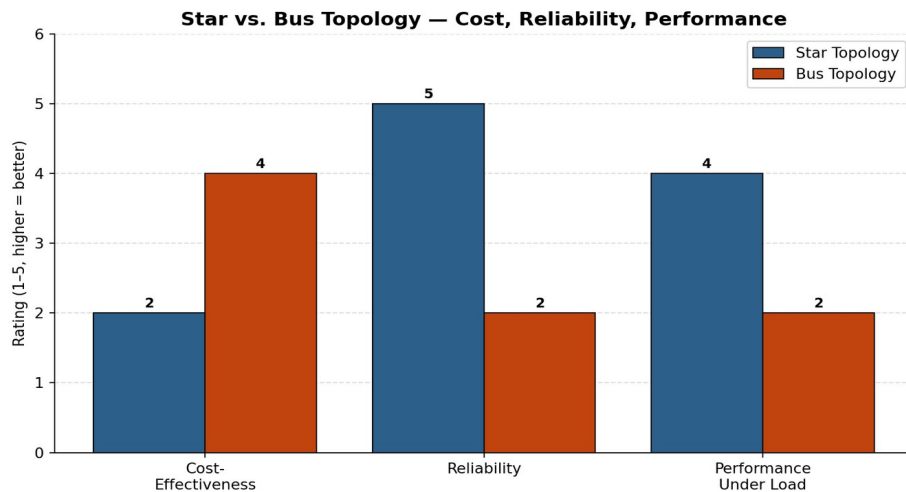


Figure 1.2 – Illustrative comparison of star vs. bus across cost, reliability, and performance.

Criterion	Star Topology	Bus Topology
Cost	Higher (switch + more cable)	Lower (single shared backbone)
Reliability	High — one bad cable isolates only one node	Low — one backbone break can down the whole segment
Performance	Stays consistent as devices are added	Degrades as more devices contend for the shared medium

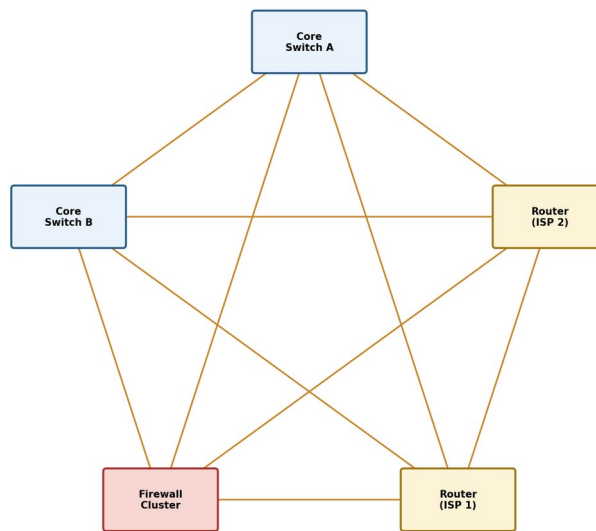
2) Mesh Topology vs. Star Topology — Fault Tolerance

Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing several alternative paths for data to travel between any two points. If one link fails, traffic can simply be rerouted through an alternate path without any disruption to overall communication — no single link or even node failure can isolate any part of a well-connected mesh.

Star topology, in comparison, has only moderate fault tolerance. While the failure of an individual node's cable does not affect any other device (as established in Question 1), the entire network depends on a single central hub or switch — if that central device fails, every connected node loses connectivity simultaneously, since there is no alternate path around it.

For this reason, mesh topology is considerably more robust and reliable in critical, uptime-sensitive environments (such as banking or data-center core networks) compared to star topology, which remains an excellent choice for smaller networks but carries this single-point-of-failure risk at its center.

Full Mesh — Bank Core Layer (Every Device ↔ Every Device)



5 devices → $n(n-1)/2 = 10$ direct links: every core device reaches every other core device in exactly one hop — no single link or node failure isolates any part of the core.

Figure 2.1 – Full mesh: every core device connects directly to every other device — no single point of failure.

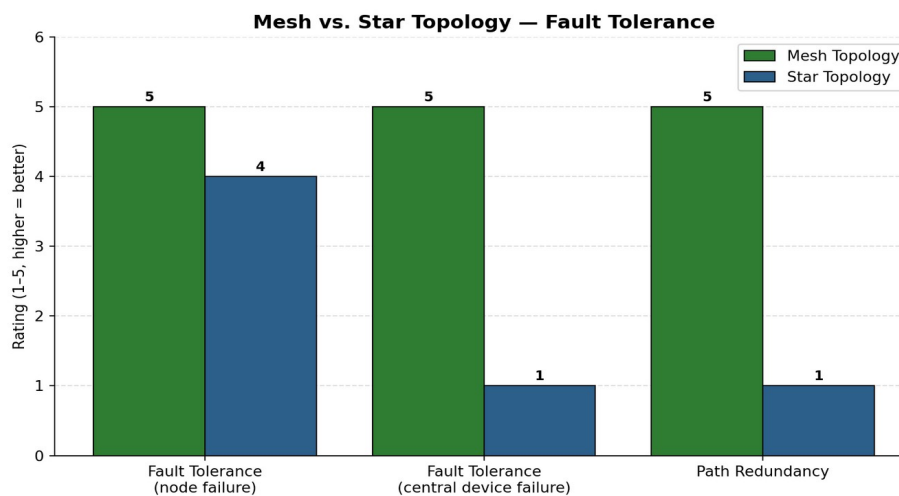


Figure 2.2 – Fault-tolerance comparison: mesh sustains both node and central-device failure; star does not survive a central-switch failure.

3) Bus Topology vs. Mesh Topology — Installation Complexity

Bus topology is simple and quick to install because it uses a single backbone cable with minimal connection points, making it well suited to small networks with only a handful of devices. Setting it up requires little more than running one cable and tapping each device into it.

Mesh topology, on the other hand, is highly complex to install: because each node must be directly connected to every other node, the number of required links grows according to $n(n-1)/2$ for n devices — for example, just 5 devices already require 10 separate connections. This complexity increases sharply as the network grows, demanding far more cabling, more ports per device, and considerably more planning and labour than the straightforward, linear setup of a bus network.

Classic Bus Topology — Coaxial Backbone (RG-58)

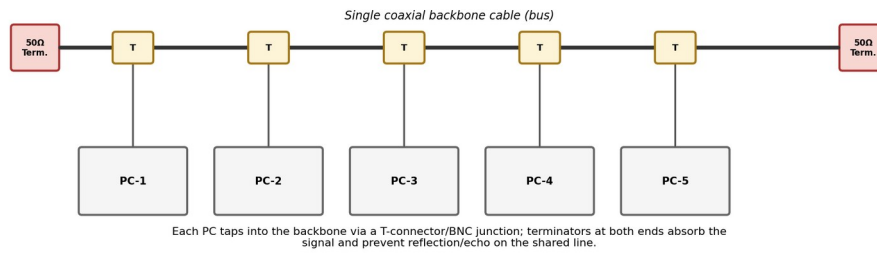


Figure 3.1 – Bus topology: a single backbone cable with minimal connection points.

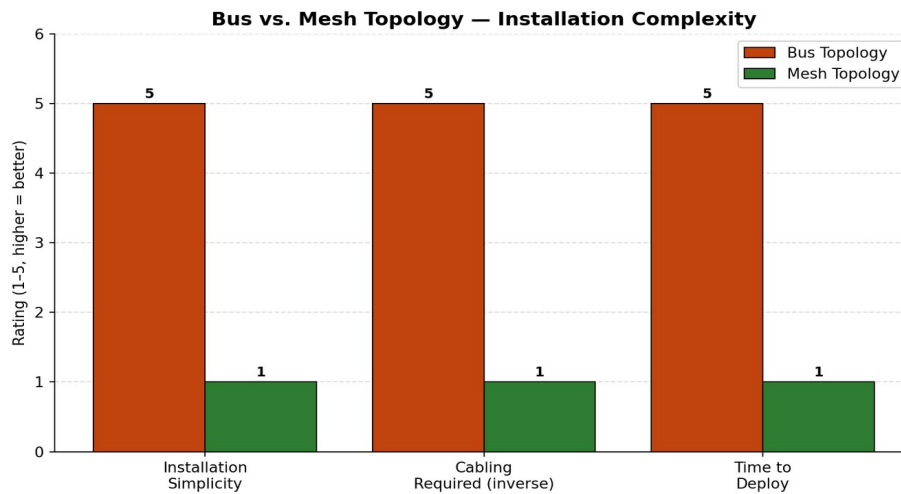


Figure 3.2 – Installation simplicity: bus topology requires far less cabling and planning than mesh.

Devices (n)	Bus — Cables Needed	Mesh — Links Needed $n(n-1)/2$
5	1 backbone cable	10 links
10	1 backbone cable	45 links
20	1 backbone cable	190 links

As this table shows, mesh cabling grows quadratically with device count while bus cabling stays constant — explaining why mesh becomes rapidly more time-consuming and costly to install as a network scales, even though it offers far superior fault tolerance (Question 2).

4) Hybrid Topology vs. Star Topology — Scalability

Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus) at different layers of the network, allowing it to expand flexibly according to organizational needs. New buildings, departments, or segments can be added by extending the appropriate layer — for example, adding a new distribution switch and star-wired access layer — without redesigning or disrupting the existing network structure.

Star topology is also scalable, but only to a limited extent: expansion is directly constrained by the port capacity of the central switch or hub. Once that device reaches its port limit, further growth requires purchasing and integrating additional switching hardware, which can introduce complexity if not planned for in advance.

Because hybrid topology allows growth to be distributed across multiple layers and combined topology types rather than depending on any single central device's capacity, it provides substantially greater long-term scalability than star topology alone — which is exactly why large organizations such as university campuses (as opposed to small offices) adopt hybrid designs.

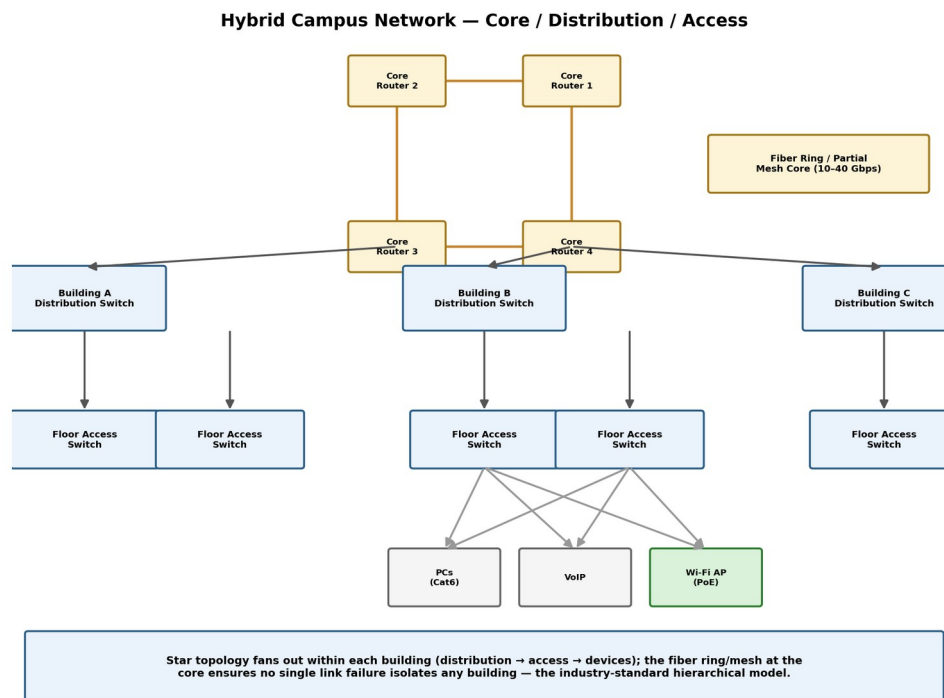


Figure 4.1 – Hybrid topology: a fiber core with star-wired distribution/access layers, scaling building by building.

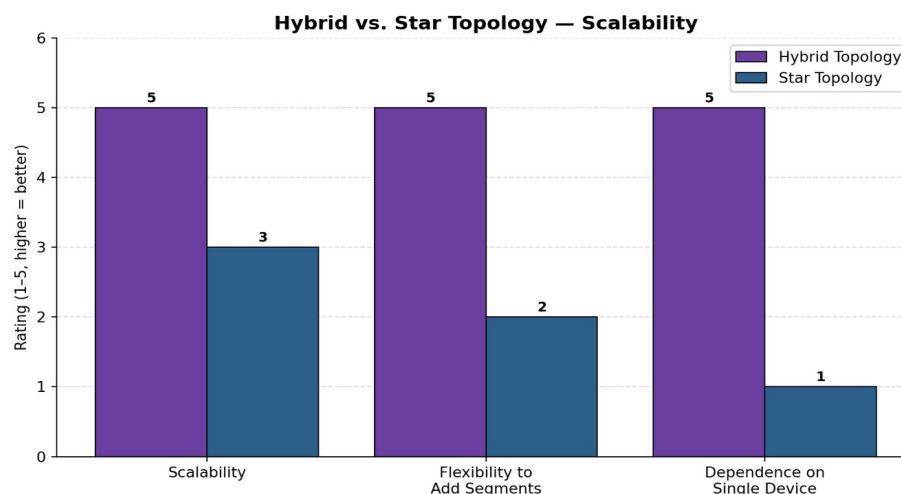


Figure 4.2 – Scalability comparison: hybrid topology is not bottlenecked by any single central device.

5) Maintenance Comparison Across All Four Topologies

In terms of maintenance, star topology is relatively easy to manage because problems can be quickly identified and isolated to a specific node or cable — a technician can check the affected link at the central switch without needing to trace the fault through other devices.

Mesh topology, while highly reliable (Question 2), is difficult to maintain due to its sheer number of connections and interdependent paths; tracing a fault or verifying every redundant link is

comparatively time-consuming, even though the network keeps running while the issue is investigated.

Bus topology is harder to maintain than star topology because a fault anywhere along the shared backbone cable can disrupt the entire segment, and locating the exact point of failure along a long cable run (without the benefit of a central device reporting per-port status) can be slow and labour-intensive.

Hybrid topology sits in between: it has moderate maintenance complexity, since it benefits from the flexibility of combining topologies, but administering multiple combined topology types at once requires careful, ongoing monitoring and a skilled network administration team who understands each layer's behaviour.

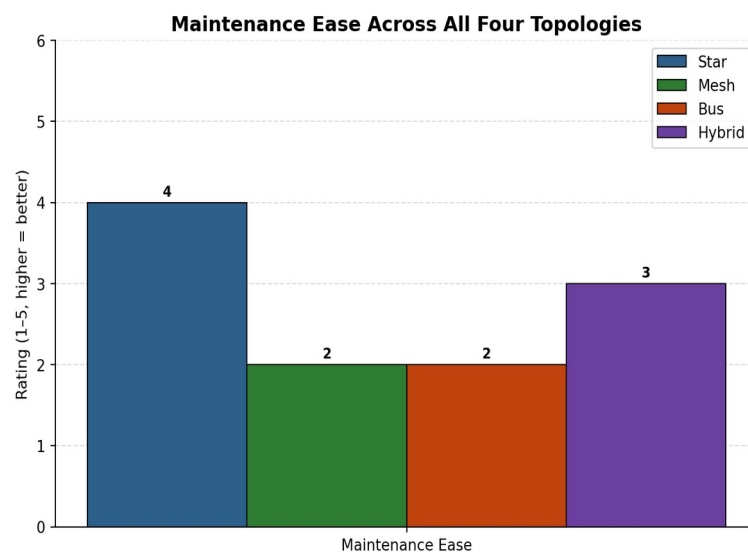


Figure 5.1 – Relative maintenance ease across star, mesh, bus, and hybrid topologies.

Topology	Maintenance Ease	Why
Star	Easy	Faults isolate to one node/cable; central switch shows per-port status
Mesh	Difficult	Very high number of interdependent links to check and trace
Bus	Difficult	Backbone faults disrupt the whole segment and are hard to pinpoint
Hybrid	Moderate	Flexible, but requires skilled administration across combined layers

Overall Summary

No single topology wins across every criterion — each represents a different trade-off between cost, reliability, performance, fault tolerance, installation effort, scalability, and maintainability. Bus topology minimizes cost and installation effort but sacrifices reliability and performance at scale. Star topology offers a strong, easy-to-maintain balance for small-to-medium networks, at the cost of depending on its central switch. Mesh topology maximizes fault tolerance and reliability for critical environments, at the cost of installation complexity and maintenance overhead that grow quadratically with network size. Hybrid topology combines these strengths — star's manageability, mesh or ring redundancy at the core, and bus-like simplicity where appropriate — making it the most scalable choice for large, multi-segment networks, at the cost of requiring more sophisticated

administration. Selecting the right topology, in practice, means matching these trade-offs to the specific scale, budget, and criticality of the network being designed.